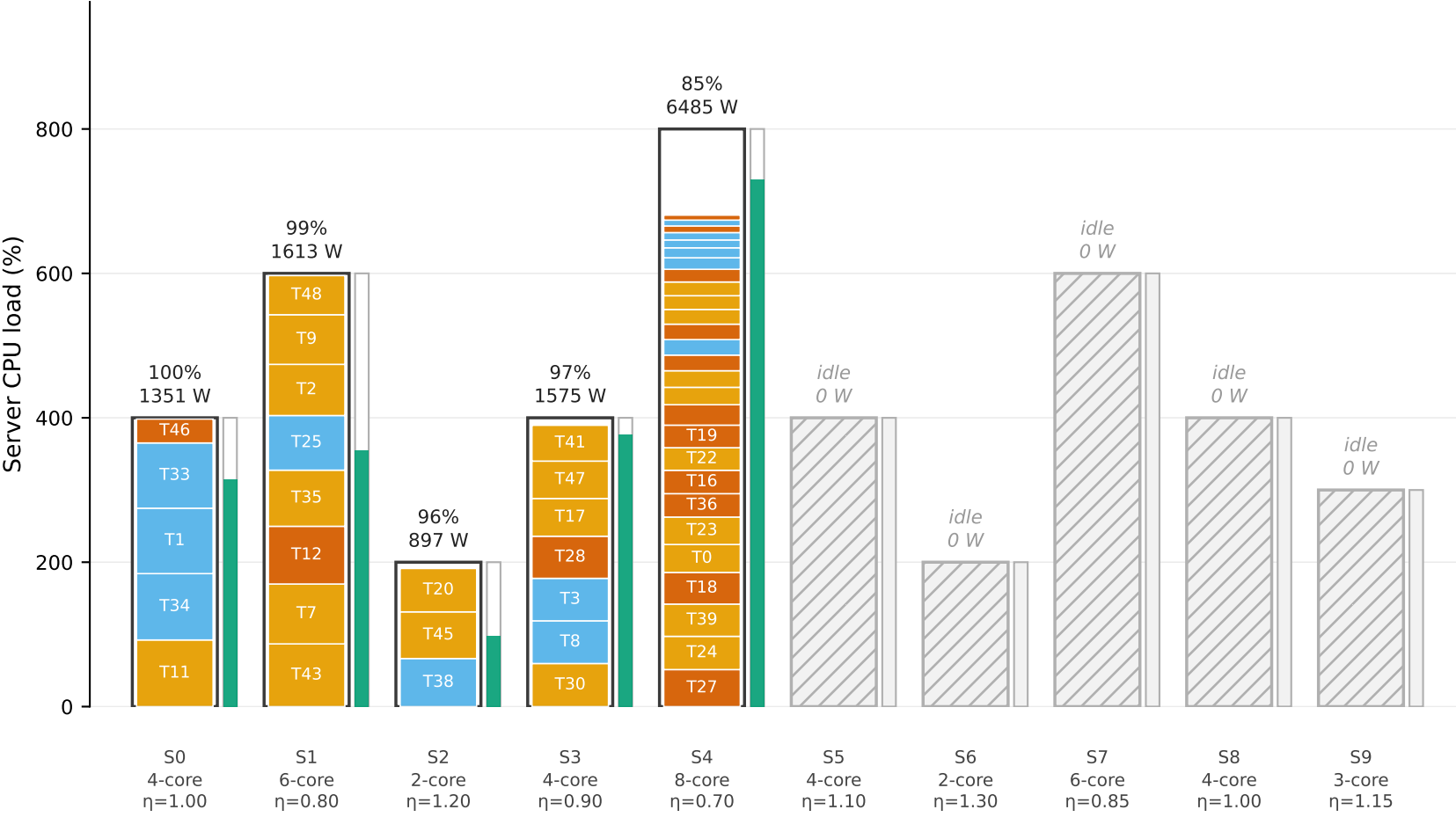


Cloud Resource Allocation — Greedy BFD (baseline) vs Simulated Annealing [balanced focus]
Simulated Annealing improves F(X) by 9.9% · 36 of 50 tasks moved

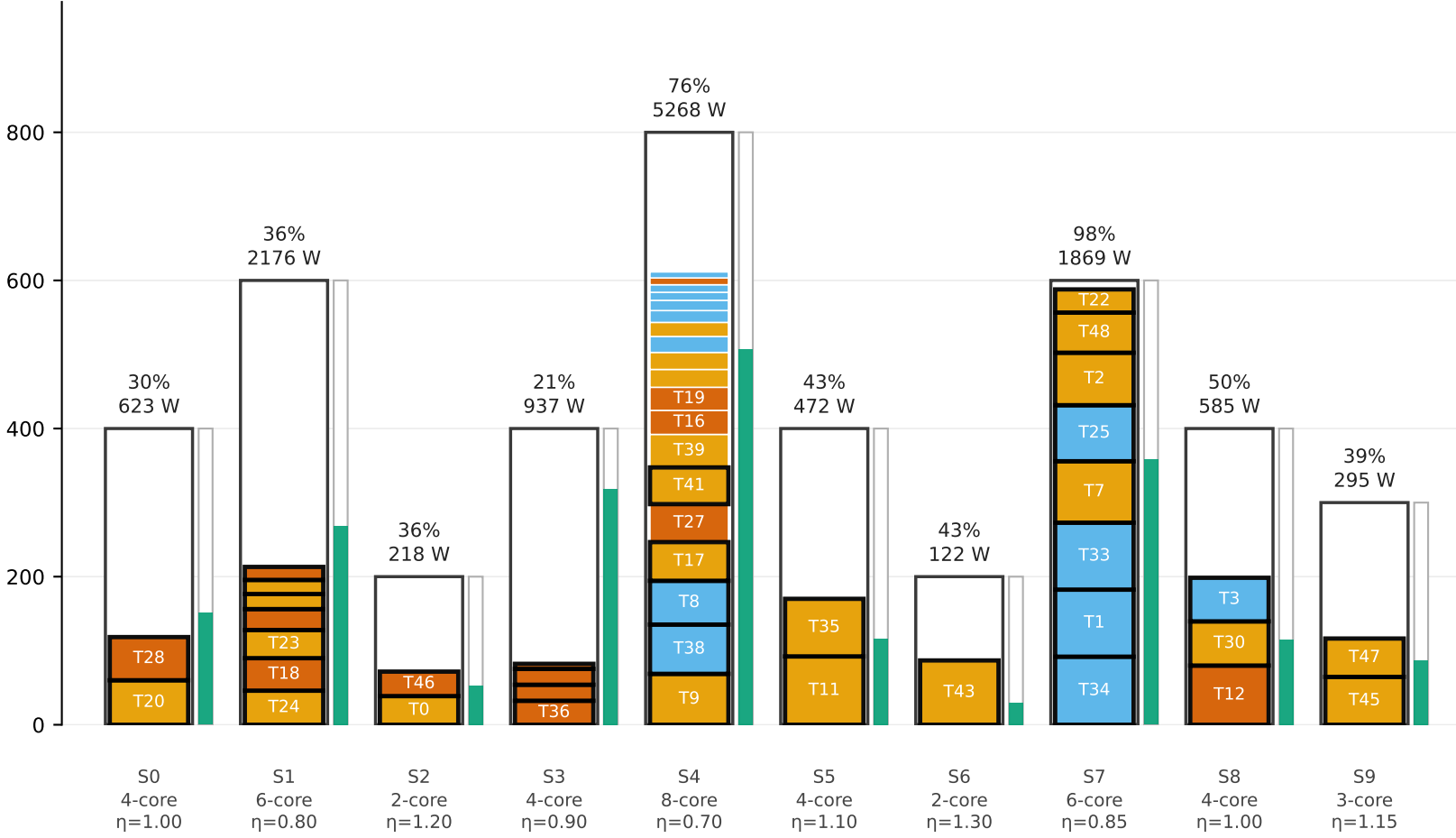
Greedy BFD (baseline)

F(X) = 2.0062 energy 11,921 W latency 30,963 ms servers 5/10 feasible



Simulated Annealing

F(X) = 1.8086 energy 12,566 W latency 23,678 ms servers 10/10 feasible



Greedy BFD (baseline) 0.95 1.06 F = 2.006

Simulated Annealing 1.00 0.81 F = 1.809

Contribution to F(X) (blue = $w_e \cdot E/E_{ref}$, magenta = $w_l \cdot L/L_{ref}$, red = penalty)

Low priority Medium priority High priority Memory gauge (fill = used / capacity) Idle server (0 W) Task on a different server than under Greedy BFD (baseline)